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Reprogramming Automatic Speech Recognition Models for Neonatal Chest Sound Separation

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***Abstract*— Stethoscope-recorded chest sounds are invaluable for providing a non-invasive, real-time assessment of heart and lung sounds. However, noisy chest sounds can affect the dependability of various algorithms that rely on clean chest sounds, making additional preprocessing necessary to isolate the desired sources and remove noise, interference, and artifacts. This paper is the first to explore the reprogramming of automatic speech recognition (ASR) models to perform neonatal chest sound separation. In particular, we reprogrammed the Whisper ASR model to perform chest sound separation. We proposed two approaches: reprogramming just Whisper's audio encoder and reprogramming the full Whisper model. Using only simple linear layers and learnable parameters, we showed that this parameter-efficient method of reprogramming Whisper effectively separates heart and lung sounds from noise on the artificial dataset. We also demonstrate the effectiveness of using the proposed method as a preprocessing step for various heart and lung sound algorithms, yielding results comparable to state-of-the-art performance. Applying the pre-trained ASR model to perform sound separation demonstrates the feasibility of efficient cross-domain model reprogramming, demonstrating the feasibility of using frozen cross-domain foundational models from a different domain on biomedical data.**

***Index Terms*— Model reprogramming, source separation, deep learning, parameter-efficient, heart sound, lung sound**

I. Introduction

CLEAN heart and lung sounds are crucial for measuring various vital parameters such as heart rate and breathing rate. The accessibility and affordability of digital stethoscopes have enabled the development of numerous algorithms that rely on cardiac or respiratory sounds. However, noisy chest sounds can compromise the performance of these algorithms, leading to errors and hindering clinical decision-making. As such, noisy chest sounds require further processing to isolate the desired sources.

One of the challenges in separating chest sounds is the difficulty and expense of obtaining pure heart or lung sounds, especially for newborns and infants. Their chest sounds are inherently more complex due to several factors: (a) Infants often cry during auscultation, making the collection process difficult. (b) Respiratory support machines and other environmental sounds are common with preterm neonates. (c) External factors, such as newborn movements, can cause the stethoscope to rub or disconnect, further contributing to noise. (d) The heart and lung sounds of neonates tend to be of higher frequency than those of adults. These complexities make effective denoising both more challenging and resource-intensive.

The challenge of limited pure heart and lung sounds has been previously addressed in several ways. One approach is to utilise statistical methods of blind source separation, such as adaptive filtering, independent component analysis (ICA), and empirical mode decomposition (EMD) [1], to separate heart and lung sounds. Some model-based source separation methods, such as non-negative matrix factorisation (NMF) [2] and efficient deep-learning-based models [3], have been proposed to separate heart and lung sounds. These model-based methods often assumed strong priors about the data, such as the sources contained in the chest sounds or the structure of the chest sounds, including the periodicity of chest sound recordings. Whilst practical, these methods are only effective if the data matches the assumptions, and generalising the model-based methods to biomedical data as a whole would be difficult.

Another approach is to use a pre-trained foundational model trained on a large dataset and fine-tune it with regularisation for downstream tasks. However, foundational models typically only exist for modalities that are easily accessible over the Internet, such as images, text, and audio. However, such foundational models are scarce in the biomedical domain. In this work, we challenge the idea of creating a foundational model for each biomedical signal domain, instead leveraging pretrained foundational models from other domains with large-scale datasets. Specifically, we explore using a model reprogramming approach to reprogram a **frozen** pre-trained foundational model initially programmed to perform speech-to-text

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transformation to perform neonatal chest sound separation as an efficient deep-learning-based supervised learning.

Model reprogramming is the repurposing of a pre-trained neural network to perform a new task without modifying the model's architecture or parameters. This is typically achieved by modifying the model's input data and/or adjusting the output interpretation by tuning the input/output transformation, rather than the internal model architecture or parameters. Model reprogramming was first proposed by Elsayed *et al.* [4], where the authors reprogrammed an ImageNet classifier to perform various tasks such as counting squares, MNIST classification, and CIFAR-10 classification. Recently, Neekhara *et al.* demonstrated cross-domain model reprogramming for various cross-domain tasks such as DNA sequence classification and sentiment classification using image classifiers [5]. This work investigates whether such cross-domain reprogramming can be extended to reprogram an automatic speech recognition (ASR) model to perform neonatal chest sound separation.

As such, this paper proposes a method for reprogramming an automatic speech recognition model to perform neonatal chest sound separation. In particular, we reprogrammed the Whisper model [6] to perform neonatal chest sound separation using three linear layers and learnable parameters, while freezing the parameters in the original Whisper model. Compared to an end-to-end supervised learning model, the proposed model utilises significantly fewer trainable parameters while still achieving comparable performance, especially in real-world heart sound benchmarks, where it outperforms previous methods.

The paper is organised as follows: Section II introduces the single-channel source separation problem and its relations to chest sound separation, model reprogramming, the Whisper model, and some common parameter-efficient fine-tuning methods. Section III then introduces the proposed chest sound separation models and the dataset used for training and evaluating the model. Section IV describes the experimental setup and the result comparison with previous methods. Section V highlights observations made about the proposed separation method. Lastly, Section VI states the limitations of the proposed work and the future research directions.

The contributions of this paper are as follows:

- We proposed two parameter-efficient approaches to reprogram a pre-trained Whisper model without the need to fine-tune the pretrained model fully.
- To the best of our knowledge, this is the first work to demonstrate the effectiveness of reprogramming an ASR model to perform neonatal chest sound separation.
- By training only three linear input/output transformations rather than fully fine-tuning the model, we reduce the cost of adaptation (total training compute, data required, etc.) for such a model.
- We demonstrate that our proposed method achieves performance comparable to or even better than previous approaches on downstream tasks, especially for heart sounds.
- We investigated the latent representation of the Whisper model in the context of neonatal chest sound separation: The role of the pre-trained weights, the optimality of the weights, and the scaling potentials of the method.

II. Background

A. Single-channel Source Separation

Single-channel source separation aims to isolate multiple source signals from a single mixture signal. Equation 1 describes a mixing process, where k source signals, $\boldsymbol{s_i}$, are combined to create a mixture signal $\boldsymbol{x}$. f describes any arbitrary mixing function. Common mixing functions include additive mixing and convolutive mixing.

$$\boldsymbol{x} = f(\boldsymbol{s_1}, \boldsymbol{s_2}, \ldots, \boldsymbol{s_k}) \tag{1}$$

The objective is to find a mapping g that reverses the mixing process and recovers the source signals $\boldsymbol{s_i}$. This is done with little to no prior knowledge of f and the source signals.

$$\boldsymbol{s_1}, \boldsymbol{s_2}, \ldots, \boldsymbol{s_k} = g(\boldsymbol{x}) \tag{2}$$

We frame the separation of neonatal chest sounds as a single-channel source separation problem. We assume that the mixture consists of chest sound recordings, which are composed of heart sounds and lung sounds. We approximate the reverse mixing process, g, with a neural network.

B. Model Reprogramming

Adversarial examples are specific inputs to a neural network designed to deceive the model yet remain indistinguishable to a human observer. This is often done by adding small perturbations to the original input. In the domain of audio, by carefully choosing the perturbation, we can induce a neural network to misclassify audio without modifying its underlying weights. Figure 1 illustrates an adversarial example where a pre-trained murmur classifier's output is altered after adding subtle noise imperceptible to human vision and audition.

Adversarial examples demonstrate that a pre-trained model's behaviour can be altered by manipulating its input without modifying its internal parameters. Elsayed *et al.* were the first to exploit this phenomenon to reprogram a model, where the authors used padded noise to reprogram a model pre-trained on ImageNet to perform various downstream tasks [4]. Notably, the reprogrammed models performed well on test datasets, suggesting that they also exhibited generalisability rather than pure memorisation of the training examples.

Since then, model reprogramming has been successfully implemented for various tasks, including but not limited to, visual reprogramming [7], [8], text reprogramming [9], or even text-to-image cross-domain reprogramming [5]. This paper introduces a novel application of model reprogramming for biomedical source separation, demonstrating its potential for isolating neonatal heart and lung sounds. Figure 2 gives a high-level overview of the proposed method.

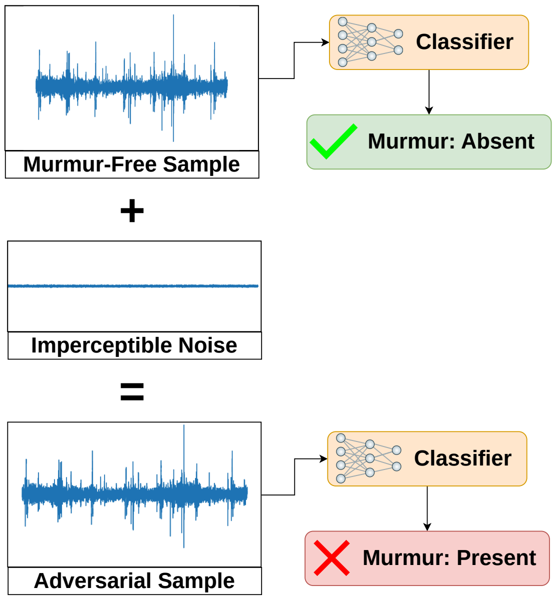


Fig. 1: An illustration of adversarial examples. Noise added to a murmur-free sample causes the classifier to misclassify as murmur, even though it remains imperceptible to humans.

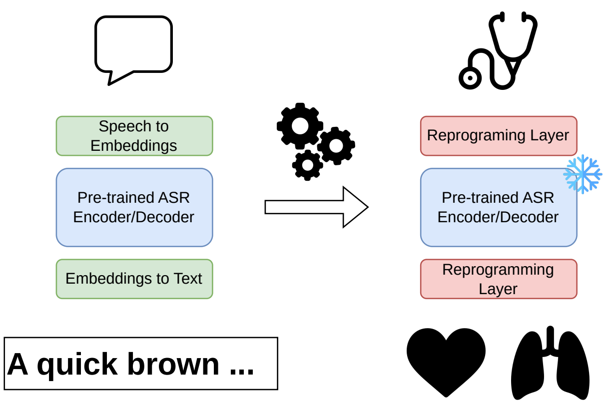


Fig. 2: A high-level description the proposed method. The proposed method changes the input and output embedding mappings while keeping the embedding backbone frozen.

C. Whisper

Whisper [6] is an automatic speech recognition (ASR) system developed by OpenAI that is trained on 680,000 hours of multilingual and multitask supervised data. The architecture employs a simple end-to-end approach, implemented as an encoder-decoder Transformer. Figure 3 shows the overall architecture of the Whisper model. The model takes a log-mel spectrogram of the input speech and passes it through the audio encoder block, which consists of two 1D convolution blocks, followed by Transformer encoder blocks. The model then produces a set of text tokens conditioned by the encoder, using the text decoder, which consists of Transformer Decoder blocks.

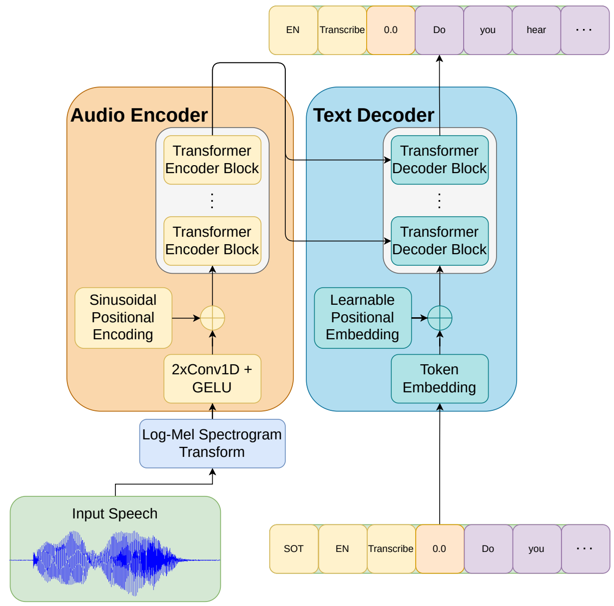


Fig. 3: Whisper model architecture. The model consists of an audio encoder and a text decoder.

D. Parameter-Efficient Fine-Tuning

Parameter-efficient fine-tuning (PEFT) is a technique for fine-tuning a pre-trained model for downstream tasks without fully updating all of the model's parameters. This is typically done by freezing most of the pre-trained model and only updating a few parameters, resulting in lower training requirements such as lower training GPU memory footprint and training samples. As such, this allows large models to be fine-tuned efficiently.

As such, this technique is popular in natural language processing, where a pre-trained large language model is fine-tuned on various downstream tasks [10]. Popular PEFT methods include Low-Rank Adaptation (LoRA) [11], Adapter Layers [12], Prefix-Tuning [13], etc. In this work, we implement linear adapter layers to align the input and output with the embedding space of Whisper.

III. Methods

A. Chest Sound Separation Model

Given a mixed chest sound $\boldsymbol{x}$, the goal of the separation model is to decompose $\boldsymbol{x}$ into k desired sources. In this case, heart sound $\boldsymbol{s}_{\text{heart}}$ and lung sound $\boldsymbol{s}_{\text{lung}}$ are the desired sources, making $k = 2$.

Inspired by the ability of foundational models to capture complex patterns, our model aims to adapt their expressive, pre-trained embeddings to separate chest sounds. For simplicity and parameter efficiency, we freeze the foundational model and use lightweight linear adapters first to transform the input audio into the pre-trained embedding space, and then remap the output sources, relying on the embedding's power rather than adding complex layers.

Figure 4 shows the architecture of the model. The model consists of three adapter layers to process the audio. The model first converts the input audio into initial embeddings using an

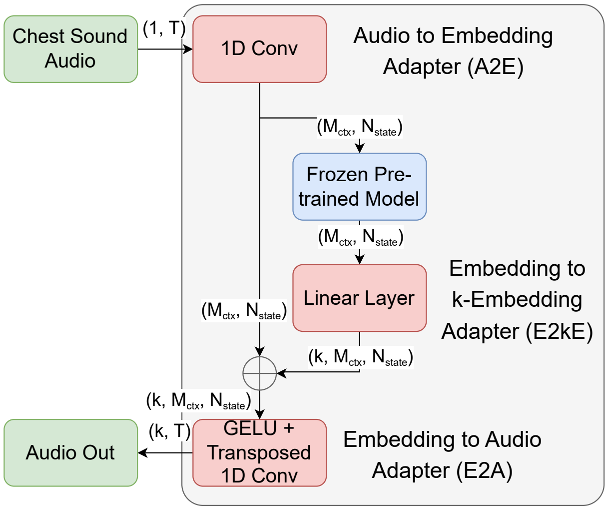


Fig. 4: Model architecture. The green blocks represent the input/output, the red blocks are tunable layers, and the blue block is the frozen pre-trained model to be reprogrammed.

audio-to-embedding (A2E) layer. The frozen pre-trained model then processes these embeddings and subsequently expands them into k-embeddings for each source by the embedding-to-k-embeddings (E2kE) layer. This processed output is added back to the initial embeddings from the A2E layer, allowing the model to add and remove information from the initial embeddings. Finally, this combined result is passed to an embedding-to-audio (E2A) layer, which converts the processed embeddings back into distinct audio tracks. We note that this proposed method can be trained with any end-to-end separation training algorithm, provided that the weights of the pre-trained model are appropriately frozen and not updated during the training process.

The A2E adapter maps chunks of audio sounds into the embedding space of the frozen pre-trained model. Given a context length l, the A2E adapter will apply a linear mapping to the audio signal chunks of length l into the embedding space of the pre-trained model with dimension N_{state}. This is implemented using 1D convolution, where the kernel size represents the context length and the number of output channels defines the embedding size. This is similar to the linear projection of image patches used in vision transformers [14], but instead of converting image patches to embeddings, we convert audio chunks into embeddings.

Similarly, the E2A adapter maps embeddings of size N_{state} back into audio chunks with context length l. This is implemented using 1D transposed convolution, where the context length of the audio chunks, l, is represented by the kernel size, and the embedding size is represented by the input channel N_{state}. The E2A adapter simultaneously decodes the heart sound and lung sound.

Lastly, the E2kE takes the embedding generated by the pre-trained model and generates k embeddings, where k represents the number of sources. This is implemented as a simple linear layer, where the input dimension is the embedding dimension N_{state}, and the output dimension is $k \times N_{\text{state}}$, and reshaped to have a shape of (k, N_{state}).

We evaluated our work using the Whisper-Base and Whisper-Tiny models [6] as the frozen pre-trained foundational models. Additionally, we reprogrammed Whisper in two ways:

1) We reprogrammed just the Audio Encoder (Whisper-Audio).
2) We jointly reprogrammed the Audio Encoder and Text Decoder (Whisper-Full).

1) Whisper-Audio: This method assumes that Whisper's Audio Encoder is available as a separate module for reprogramming. We removed the two 1D convolutions before the transformer encoder blocks and replaced them with the A2E adapter. Finally, we use the cross-attention outputs from the Audio Encoder as the encoded chest sounds and attach the E2kE layer for audio source separation tasks.

For this implementation, we did not pad the 10-second segment audio to 30 seconds, as with Whisper's training, since we did not find any performance gain from doing so, while adding additional computational requirements.

2) Whisper-Full: This method jointly reprograms Whisper's audio encoder and text decoder for audio source separation. Figure 5 shows the information flow from the A2E layer to the E2kE layer via the frozen Whisper model.

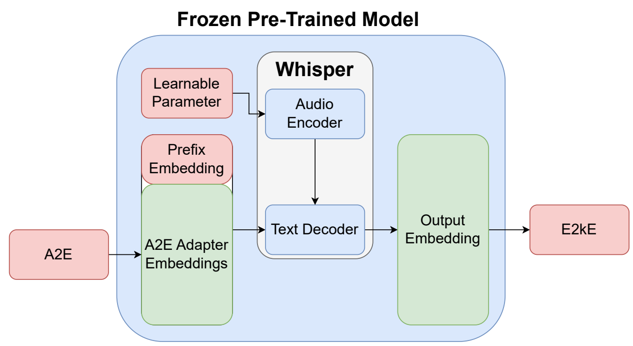


Fig. 5: Information flow from A2E layer to E2kE layer via frozen Whisper model. Additional tunable parameters (coloured red) are added.

To reprogram the audio encoder, we learn a global spectrogram representation as the input to the audio encoder to condition Whisper's text decoder. This is done by using a tunable parameter of shape $(N_{\text{Mel}}, M_{\text{Frame}})$, the expected input shape of the audio encoder. This parameter is initialised with all zeros, and the model learns the representation throughout the training process.

To reprogram the text decoder, we propose the idea of passing the text embeddings generated by A2E into the text decoder. We concatenate a prefix and the embeddings from the A2E adapter to form the prompt for Whisper's text decoder. The decoder produces a set of output embeddings and is fed to the E2kE layer. We first initialised the prefix with {<|startoftranscript|>, <|en|>, <|transcribe|>, <|notimestamp|>} tokens. This prefix embedding can then be fine-tuned during the training phase, similar to soft prompt tuning [15].

B. Datasets

We sourced our training and testing datasets from the Monash Children's Hospital dataset and the CirCor DigiScope Dataset [16]. A portion of the Monash Children's Hospital dataset was used for training the proposed models, with the remainder reserved for evaluating their performance. The CirCor DigiScope Dataset was used as an additional dataset for performance evaluation.

1) Monash Children's Hospital Dataset - Pure Source Dataset: We trained our proposed model on chest sounds generated by artificially mixing pure heart, lung, and noise sources and evaluated the proposed model's separation performance on a separate testing dataset. We used the pure source from [2] to create our artificial mixtures. These sources were obtained from newborn babies admitted to Monash Children's Hospital. These noises were broken down into general noises and respiratory support noises. General noise includes crying and stethoscope movement (STMV) noise, such as rubbing and disconnecting noise. Respiratory support noise includes ventilator continuous positive airway pressure (CPAP) noise and bubble CPAP noise. The data was collected with approval from the Monash Health Human Research Ethics Committee (HREA/18/MonH/471).

Clean heart sounds, s_{heart}, clean lung sounds, s_{lung}, and noise sources s_{noise} were first normalised to have equal signal power. They were then mixed additively (Equation 3) and convolutively (Equation 4) to form the mixture, x_{mix}, where $*$ is the convolution operation. α, β, and γ were chosen based on the desired signal power in the mixture. During training, these values were randomly selected from a range of -10 dB to 10 dB, whereas in the testing dataset, the values were fixed to be {-10 dB, -5 dB, 0 dB, 5 dB, 10 dB}. For convolutive mixtures, random finite impulse response (FIR) filters for heart, a_{heart} and lung, a_{lung}, were added to mimic small echoes that the stethoscope could capture. Table I shows the total number of combinations from each training partition.

$$x_{\text{mix, add}} = \alpha s_{\text{heart}} + \beta s_{\text{lung}} + \gamma s_{\text{noise}} \tag{3}$$

$$x_{\text{mix, conv}} = a_{\text{heart}} * \alpha s_{\text{heart}} + a_{\text{lung}} * \beta s_{\text{lung}} + \gamma s_{\text{noise}} \tag{4}$$

Initial testing revealed that including STMV noise in the training set drastically hinders model performance, while excluding no-noise cases slightly improves it. Note that these cases are still present in the testing set used for evaluation.

2) Monash Children's Hospital Dataset - Synchronous Vital Signs Dataset: In addition to the pure source dataset, 33 chest sounds were collected along with synchronous vital signs. These data included second-by-second heart rate data collected using electrocardiogram (ECG) sensors and breathing rate data collected using impedance tomography sensors. The dataset is subdivided into 12 recordings captured in the presence of respiratory support noise and 21 recordings without respiratory support noise. This data was kept for testing purposes and was not used in training.

TABLE I: Training, validation and testing dataset. The dataset was split patient-wise following [17]. *Combination excludes the STMV noise and multiple SNR steps, which are dynamically generated at training time. See section IV-A.2

Source	# of recordings		
	Training	Validation	Testing
Heart	10	2	5
Lung	6	2	1
Bubble CPAP	2	3	2
Ventilator CPAP	4	2	2
Cry	31	6	5
STMV	12 (unused)	4	2
Total combinations	*2220	3000	2750

3) CirCor DigiScope Dataset: The CirCor DigiScope Dataset [16] contains heart sounds from various age groups, from neonatal to young adult. The data were collected from 2014 to 2015 as part of two mass screening campaigns in Paraiba, Brazil, using a Littmann 3200 stethoscope. The digital stethoscope was placed at one of the four main auscultation points, located around the aortic, pulmonary, tricuspid, or mitral valves. The heart signals were acquired at a sampling frequency of 4 kHz with a 16-bit resolution.

Each stethoscope recordings contain reference heart segmentation annotations. The recordings were first segmented automatically using the three algorithms by [18]–[20]. The annotated segmentations were then provided to two cardiac physiologists, who independently analysed the recordings. The experts then randomly selected one of the annotated segmentations, and the annotation file was included in the dataset if the experts agreed with the annotations. If they disagreed, the experts selected another annotated segmentation. If the expert disagreed with all the annotated segmentations, the expert manually annotated at least five heartbeat cycles.

TABLE II: Age group breakdown of the CirCor Dataset

Age group	# participant	# recordings
Neonate	6	10
Infant	126	269
Total	132	279

Since the separation methods were only trained on neonates, we filtered the dataset to include only neonates and infants. This is done because the statistical properties of heart sounds differ substantially among each age group. Additionally, since the segmentation annotations are only applied to recordings with distinct heart sounds, we incorporated breathing sounds and external noise from the testing partition of the Monash Children's Hospital dataset. Table II shows the dataset breakdown used.

IV. Experiments and Results

A. Experimental Setup

1) Model Parameters: Table III summarises the context size used for each proposed model, the number of tunable parameters, and the total number of parameters. We note that the parameter count for Whisper-Full differs from the original implementation due to the exclusion of parameters related to the token embeddings in the text decoder.

TABLE III: Parameters used for each model.

Model	Context Size l	Tunable Parameters	Total Parameters	Percentage Tunable
Whisper-Audio-Tiny	192	444k	8.1M	5.5%
Whisper-Audio-Base	192	722k	20.5M	3.5%
Whisper-Full-Tiny	192	683k	18.0M	3.8%
Whisper-Full-Base	256	1.03M	46.3M	2.2%

2) Data Augmentations: The following augmentations were applied to the training dataset:

1) The source signals were randomly inverted with a 50% probability.
2) The source signals were randomly cropped to have 32,000 samples, corresponding to 8 seconds with a 4 kHz sampling rate.
3) The source signals were randomly scaled to have a signal-to-noise ratio between -10 dB and 10 dB.

3) Method Comparison: We compared the proposed methods with the following methods:

- **Singular Spectrum Analysis (SSA) [21]**: SSA decomposes the signal into principal components. The top eigenvector pairs of components with frequencies less than 250 Hz are assigned as heart sounds. Modifications for neonates were implemented following [2].
- **NMF** [2]: A non-negative matrix factorisation method of decomposing the non-negative spectrogram into an activation matrix and a basis matrix. During the training phase, the basis vector is learned. During the evaluation phase, the learned basis vector is used to decompose the unseen spectrogram.
- **NMCF** [2]: Similar to NMF, but the basis vector is jointly optimised during inference time.
- **NeoSSNet** [17]: A deep-learning-based method of separating neonatal chest sounds into heart and lung sounds. It consists of convolution layers and transformer layers.

B. Signal Distortion Measurements

To assess the quality of sources separated by different methods, we utilised the signal-to-distortion ratio (SDR) [22] and scale-invariant signal-to-distortion ratio (SI-SDR) [23]. Table IV shows the median SDR and SI-SDR improvement achieved by various neonatal chest sound separation methods on the Monash Children's Hospital pure source testing dataset. These improvements are calculated by comparing the separated sources to the original samples. A higher value signifies greater improvement and lower distortion.

None of the proposed models achieves state-of-the-art performance for heart sounds, yielding results comparable to those of the NMF and NMCF methods. In contrast, the model shows significant promise for separating lung sounds. The proposed Whisper-Audio-Tiny model outperformed the state-of-the-art separation model in the presence of respiratory

TABLE IV: Median SDR and SI-SDR improvement for heart and lung sources measured in dB. The best and second-best results are highlighted in **boldface** and underlined, respectively.

Heart	No Noise		General Noise		Resp Support Noise	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
NeoSSNet	**16.70**	**16.12**	**19.64**	**19.32**	**11.50**	**11.64**
NMF	14.64	14.23	17.42	17.60	7.04	7.30
NMCF	14.85	14.36	17.57	17.80	8.67	8.53
SSA	-1.64	-7.29	-9.07	0.41	0.68	-2.06
Whisper-Audio-Tiny	14.80	13.92	16.67	16.29	9.97	10.31
Whisper-Audio-Base	15.15	14.49	16.96	16.49	9.71	9.96
Whisper-Full-Tiny	14.87	14.09	16.67	16.18	9.37	9.44
Whisper-Full-Base	14.58	13.58	16.64	16.02	9.22	9.79
Lung	**No Noise**		**General Noise**		**Resp Support Noise**	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
NeoSSNet	**13.49**	**12.40**	**12.47**	**12.09**	14.35	13.79
NMF	6.14	5.40	9.81	10.19	11.02	10.79
NMCF	3.07	0.74	9.72	9.12	11.68	10.93
SSA	3.98	3.04	0.55	0.49	5.99	4.73
Whisper-Audio-Tiny	12.92	11.39	7.84	8.14	**15.02**	**14.26**
Whisper-Audio-Base	11.67	10.37	8.11	8.06	14.42	13.67
Whisper-Full-Tiny	10.55	8.81	10.19	9.97	14.21	13.52
Whisper-Full-Base	9.36	8.01	9.08	9.10	13.01	12.48

support noises. This is followed closely by the Whisper-Audio-Base model. Lastly, the Whisper-Full family of models comes in last but can still outperform the previous NMF and NMCF methods in pure heart-lung mixtures and mixtures that contain respiratory support. Interestingly, apart from the Whisper-Tiny model, all proposed models performed worse in the general noise category than previous methods.

We verify these performance claims by visually inspecting the estimated sources against the ground-truth sources. Figure 6 shows the estimated sources separated by Whisper-Audio-Tiny on a particular chest sound with general noise.

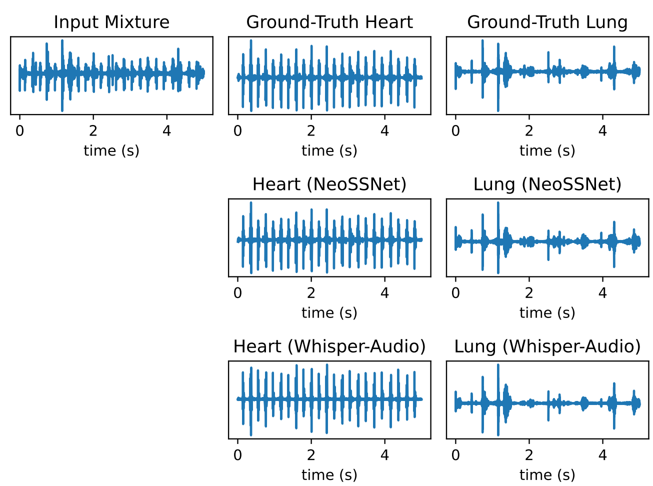


Fig. 6: Heart and lung estimates from the Whisper-Audio-Tiny model compared to NeoSSNet estimates and the ground-truth sources.

Here, we observe that the Whisper-Audio-Tiny estimated source is relatively close to the ground-truth sounds. Interestingly, Whisper-Audio-Tiny exhibits relatively flat background noise heart pulses. In contrast, the heart sound estimated by NeoSSNet contains reverberations similar to the ground-truth heart sounds. When inspecting the separated lung sounds, we note that Whisper-Audio-Tiny missed a breath at the two-second mark, which was captured by NeoSSNet. These imperfections caused the estimated Whisper-Audio-Tiny model to lag behind NeoSSNet. Nonetheless, Figure 6 shows that the proposed separation method can effectively separate the chest sound, as evidenced by the visually distinct main heart and breathing peaks in the estimated sources.

While none of the proposed models can definitively outperform NeoSSNet, with performance comparable to NMF and NMCF methods, the fact that the reprogrammed model achieves similar performance to NeoSSNet demonstrates the impressiveness of cross-domain reprogramming an ASR model for chest sound separation.

C. Heart Rate and Breathing Rate Estimation

We evaluated our proposed separation methods as a denoising algorithm for heart and breathing rate estimation using the Monash Children's Hospital synchronous vital sign dataset. For heart rate estimation, we adapted the Springer *et al.* [18] segmentation algorithm, modifying the heart rate range to 110-200 beats per minute [24], and estimated heart rate from detected S1 and S2 peaks. Breathing rate was estimated using a 300-450 Hz power spectral envelope and peak detection within a 15-100 breaths per minute range [24]. We compared the performance of these estimation algorithms against ground truth synchronous data from external sensors. We report the mean improvements and standard deviations.

Table V presents the algorithmic improvements for heart and breathing rate estimations (measured in beats per minute and breaths per minute). A larger number indicates a greater improvement and lower error rate.

TABLE V: Mean heart and breathing rate improvement. The best and second-best results are highlighted in **boldface** and underlined, respectively. *Nil = No Respiratory Support noise. **Resp = Respiratory Support noise.

Heart	Nil*		Resp**	
	Mean	Std	Mean	Std
NeoSSNet	2.12	11.11	3.54	16.43
NMF	1.97	12.73	2.75	14.77
NMCF	1.63	12.06	<u>4.68</u>	19.45
Whisper-Audio-Tiny	2.23	10.36	2.61	17.01
Whisper-Audio-Base	2.21	9.54	2.52	14.91
Whisper-Full-Tiny	<u>2.33</u>	9.63	3.41	16.80
Whisper-Full-Base	**2.96**	11.81	**4.87**	17.43
Lung	**Nil***		**Resp****	
	Mean	Std	Mean	Std
NeoSSNet	1.50	8.71	<u>1.11</u>	5.73
NMF	**1.57**	10.50	-0.63	7.43
NMCF	<u>1.51</u>	11.31	-1.83	8.07
Whisper-Audio-Tiny	1.06	7.18	0.17	4.28
Whisper-Audio-Base	0.07	7.68	-0.07	5.40
Whisper-Full-Tiny	1.35	9.30	**1.34**	8.59
Whisper-Full-Base	0.75	7.10	-0.55	5.92

Whisper-Full-Base is the most effective model for improving heart rate estimation. Whisper-Full-Tiny, Whisper-Audio-Tiny, and Whisper-Audio-Base follow it. Compared to previous methods, the proposed method outperforms all previous methods in the absence of respiratory support while being competitive in the presence of respiratory support.

Except for Whisper-Full-Tiny, the proposed separation methods generally perform poorly for lung sounds. Notably, Whisper-Full-Tiny closely matches the previous state-of-the-art model, even surpassing it when respiratory support noise is present. Whisper-Audio-Tiny closely follows Whisper-Full-Tiny but achieves no significant improvement with respiratory support noise. Whisper-Audio-Base sees no improvement in either case. Conversely, Whisper-Full-Base sees some improvement when there is no respiratory support noise, at the cost of some performance regression when there is respiratory support noise. These results are surprising, given that the proposed separation algorithm performed well in lung sound separation for signal distortion measurements in the presence of respiratory support noise. Despite these limitations, the proposed methods still performed well compared to NMF and NMCF in the presence of respiratory support.

D. Heart Segmentation

We evaluate the proposed separation method as a denoising algorithm for heart segmentation. We used the heart segmentation method developed by Springer *et al.* [18], adjusting the heart rate range to be between 110-200 bpm for neonates and 80-190 bpm for infants. To evaluate performance, we calculate F1, precision, and recall scores based on true positive, false negative, and false positive conditions (detailed in Table VI).

TABLE VI: Classification outcomes for heart segmentation.

Conditions	Description
True Positive (TP)	Midpoint of the algorithmic segmentation ($S1_{est}$, $S2_{est}$) and the midpoint of the reference segmentation ($S1_{truth}$, $S2_{truth}$) is within 75ms of each other
False Positive (FP)	Midpoint of the algorithmic segmentation ($S1_{est}$, $S2_{est}$) is predicted but midpoint of the reference segmentation ($S1_{truth}$, $S2_{truth}$) is not within 75ms of each other
False Negative (FN)	Midpoint of the reference segmentation ($S1_{truth}$, $S2_{truth}$) is present but midpoint of the algorithmic segmentation ($S1_{est}$, $S2_{est}$) is not within 75ms of each other

We applied the audio segmentation algorithm to the modified CirCor DigiScope dataset, both before (baseline) and after separation, using various methods. Since the DigiScope chest sound recordings were relatively clean, yielding high baseline performance, we augmented the dataset by adding breathing sounds and environmental noises from the Monash Children's Hospital test set. Table VII shows the performance of each separation method when used as a denoising algorithm for heart sound segmentation.

All proposed and previous methods enhance the segmentation algorithm and demonstrate some generalisation to distribution shifts for heart sound. This is evident by their effective processing of heart sounds from the CirCor DigiScope dataset, which originates from a distinct neonate and infant population.

TABLE VII: Segmentation results for the various separation methods. The best and second-best results are highlighted in **boldface** and underlined, respectively. The baseline method indicates the segmentation performance before separation.

Methods	S1			S2		
	F1	Precision	Recall	F1	Precision	Recall
Baseline	42.81	48.29	38.44	41.15	45.87	37.32
NeoSSNet	61.98	62.25	<u>61.70</u>	<u>64.38</u>	64.11	<u>64.64</u>
NMF	60.15	61.08	59.25	62.94	63.29	62.59
NMCF	59.99	60.18	59.79	63.80	63.50	64.11
Whisper-Audio-Tiny	**63.06**	**64.07**	**62.07**	**65.45**	**65.94**	**64.96**
Whisper-Audio-Base	61.57	62.92	60.27	63.32	64.04	62.62
Whisper-Full-Tiny	<u>62.46</u>	<u>63.62</u>	61.33	64.30	<u>64.87</u>	63.73
Whisper-Full-Base	60.06	60.86	59.27	63.72	63.92	63.53

However, the result still does not show complete robustness to out-of-distribution generalisation as the noise and lung sounds introduced were from the Monash Children's Hospital testing dataset.

The proposed model is competitive with and often outperforms previous methods. Interestingly, the heart segmentation performance improvement is consistent with the heart rate estimation in the previous section, but not the median SDR and SI-SDR metrics. Since the heart segmentation and heart rate estimation datasets were not used for training, this suggests that the proposed reprogrammed model generalises better under heart sound distribution shifts, where the heart sounds come from a completely different dataset compared to previous methods.

We observe that the Whisper-Audio models consistently outperform the Whisper-Full models, and the tiny models surpass the base models. This result suggests that smaller models perform better than larger models, indicating a relationship between model size and model robustness to reprogramming.

E. Computational Time

We measured the computational time for each method to process a 10-second sample in the testing set. We run the evaluation algorithm 100 times and calculate the mean and standard deviation of the time for each method. All methods were tested on a system with an Intel Core i5-14600K processor.

For the Whisper-Full reprogramming method, we note that the output of the audio encoder can be cached as it remains constant for any given input. We report this optimisation implementation in Table VIII for Whisper-Full.

As shown in the table, caching the audio encoder output significantly reduces the computation time for the Whisper models. The Whisper-Full-Base model saw a time reduction from 273.8 ms to 79.0 ms, while the Whisper-Full-Tiny model's time decreased from 135.0 ms to 40.0 ms.

Among all, our previous method, NeoSSNet, was the most efficient, with an average processing time of just 11.7 ms. Our proposed Whisper-Tiny-Audio method was also highly competitive, achieving a latency of 18.8 ms. In contrast, the NMF and NMCF methods, which rely on iterative gradient descent for matrix factorisation, were considerably slower.

TABLE VIII: Mean computational time and standard deviation (STD) time for each method. **Boldface** and underline indicate the lowest and second lowest computation time, respectively. *Methods were evaluated on MATLAB R2024a

Method	Mean time	STD time
NeoSSNet	**11.7 ms**	1.4 ms
Whisper-Audio-Tiny	<u>18.8 ms</u>	5.6 ms
Whisper-Audio-Base	48.2 ms	5.8 ms
Whisper-Full-Tiny	40.0 ms	6.2 ms
Whisper-Full-Base	79.0 ms	5.8 ms
NMF*	187.2 ms	2.9 ms
NMCF*	10.1 s	0.092 s

V. DISCUSSION

A. Whisper-Base vs Whisper-Tiny

Analysis of the three experiments reveals that the reprogrammed Whisper-Tiny models generally outperform the Whisper-Base models. This finding holds for both the full model and audio-encoder-only configurations, with the sole exception of the heart rate estimation metric. We propose two primary conjectures to explain this outcome:

1) **Greater Model Complexity:** The Whisper-Base model's larger architecture, with more parameters and layers, demands a substantial amount of data for effective reprogramming. Our relatively small training dataset is likely insufficient to fully leverage the learning capacity of this more complex model, leading to suboptimal performance compared to the more data-efficient Tiny model.
2) **Over-Specialisation to the Source Task:** The Whisper-Base model may be highly optimised for its original purpose of Automatic Speech Recognition (ASR). This specialisation can result in learned features that are too specific to ASR, making them less adaptable to dissimilar, cross-domain tasks, such as source separation. This "negative transfer" hinders the model's ability to be effectively reprogrammed.

Our results also echo the findings from [25] that the "bigger is better" notion does not always apply, especially to out-of-domain reprogramming, at least in the context of federated learning. The authors used various sizes of ResNet and ResNeXT models pretrained on ImageNet to perform classification on the CIFAR-10, Oxford-IIIT, and Blood-MNIST datasets. Interestingly, CIFAR-10 and Oxford-IIIT datasets benefited from model scaling, while Blood-MNIST observed performance regression as model size increased.

At the same time, the findings from [26] showed that the "bigger is better" notion may be true some of the time for cross-domain reprogramming. Particularly, cross-domain reprogramming scales on the Devign dataset and Big-Vul dataset in terms of F1 score, but not for the D2A dataset.

These observations contribute to a more nuanced understanding of scaling cross-domain knowledge transfer, and that the size of the base model alone does not guarantee better performance. This highlights the need for further investigation into the complex relationship between the model size and performance in the context of cross-domain reprogramming.

B. Whisper-Full vs Whisper-Audio

As noted in the section on computational time, the output of Whisper-Full's audio encoder is independent of the input and can be cached. This means that reprogramming of the Whisper-Full model is limited to reprogramming its text decoder, which simplifies the comparison to a direct contrast between reprogramming the audio encoder and the text decoder.

Analysis of the three experiments indicates that reprogramming the audio encoder is a more effective strategy for enhancing model performance than adapting the full Whisper model. This leads to the inference that Whisper's audio encoder is more pliable to reprogramming than its text decoder. We posit the following hypotheses:

1) **Greater Model Complexity:** The reprogramming of the complete Whisper model entails a multi-faceted optimisation problem, necessitating the identification of optimal parameters for the adapter features, audio encoder conditioning, and prefix embedding.
2) **Domain Alignment:** The audio encoder and text decoder operate in fundamentally different feature spaces. Aligning such distinct representations requires a complex adaptation, such as a deep translation layer, rather than the simple linear transformations provided by our adapters [27].
3) **Robustness of Text Decoder:** The architectural design of Whisper is characterised as a "robust decoder, weak encoder." When presented with degraded audio signals, the audio encoder produced different conditional embeddings, while the text decoder was still able to decode spoken words from the given audio embeddings. As such, the text decoder is more resistant to adversarial examples and, consequently, more resistant to changes in its output from adversarial inputs.

Therefore, when adapting the Whisper model to a novel audio dataset, prioritising the adaptation of the audio encoder is the recommended initial strategy. However, for other classes of biomedical signals, such as electrocardiogram (ECG) data, the optimal approach—whether aligning with audio or text embeddings—remains an open question. The selection of a strategy is contingent upon the intrinsic properties of the signal and its correspondence with the model's existing modalities, warranting further investigation to ascertain the most efficacious adaptation methodology.

To better capture complex relationships between audio segments and text embeddings, we made the model's audio-to-embedding (A2E) and embedding-to-audio (E2A) layers deeper. We achieved this by adding two MBConv blocks [28] after the A2E layer and two more before the E2A layer.

As shown in Table IX, this deeper model significantly outperforms the simpler linear mapping model, with performance approaching the fully supervised NeoSSNet model. This result confirms that the model benefits from a more complex method of mapping audio embeddings to text embeddings.

TABLE IX: Performance of the model after having MBConv blocks compared to the base model and NeoSSNet.

Heart	No Noise		General Noise		Resp Support Noise	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
Whisper-Full-Tiny	14.87	14.09	16.67	16.18	9.37	9.44
+ MBConv blocks	16.14	15.92	18.63	18.30	10.76	10.89
NeoSSNet	**16.70**	**16.12**	**19.64**	**19.32**	**11.50**	**11.64**

Lung	No Noise		General Noise		Resp Support Noise	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
Whisper-Full-Tiny	10.55	8.81	10.19	9.97	14.21	13.52
+ MBConv blocks	13.25	12.10	**13.31**	**12.46**	**15.04**	**14.34**
NeoSSNet	**13.49**	**12.40**	12.47	12.09	14.35	13.79

However, this improvement came with a significant drawback. The addition of the four MBConv blocks increased the number of tunable parameters to 6.6 million. This is only 2.2 million fewer than the fully tunable NeoSSNet model. Such a significant increase in parameters undermines the primary purpose of model reprogramming.

C. Training Sample Diversity

Given the parameter efficiency of our proposed model, with the largest variant requiring only 1 million tunable parameters for chest sound separation, we investigated its performance as a function of training dataset size. For this analysis, we utilised the Whisper-Full-Tiny model, which contains 683k tunable parameters. Figure 7 summarises the relationship between the number of heart and lung training samples and key performance metrics.

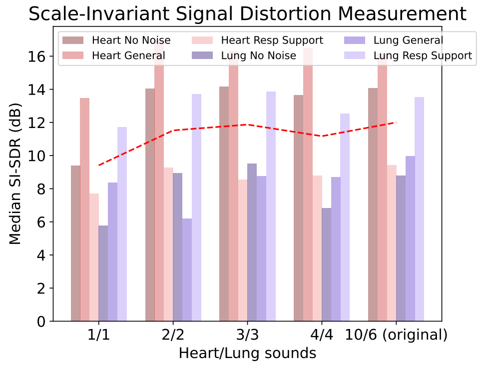
As shown in Figure 7a and 7b, both the SI-SDR and the synchronous vital sign metric began to plateau after only two heart/lung training pairs. The minor performance dip for the vital sign metric at four pairs can be attributed to the run-to-run variance of the model training process.

In contrast, the heart segmentation metric showed a clear and positive correlation with dataset size (Figure 7c). Performance on this task consistently improved as more training pairs were added. We propose two complementary hypotheses for this discrepancy:

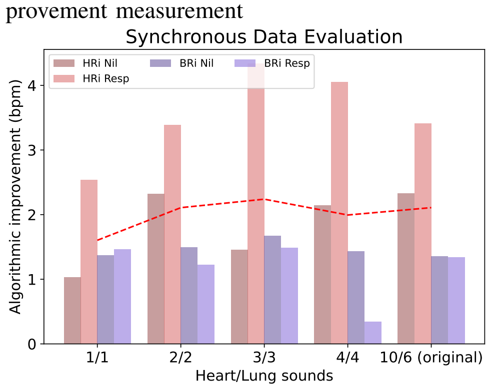
1) **Metric Sensitivity:** The heart segmentation evaluation is a measure of local separation quality, whereas SI-SDR and the vital sign metric assess global performance. Consequently, the segmentation metric is more sensitive to subtle imperfections in the separated audio that can be improved with more data.
2) **Generalisation to Distributional Shift:** The CirCor dataset, used for evaluation, exhibits a known distributional shift from our training data in terms of patient populations and recording equipment. A model trained on a larger dataset is better equipped to generalise across these variations, which is more readily captured by the sensitive local segmentation task.

D. Pre-trained vs Random Weight Initialisation

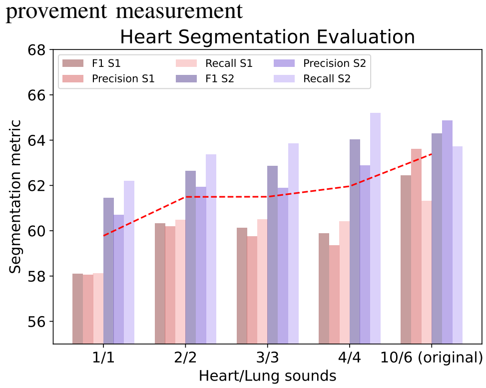
To determine the effectiveness of Whisper's pre-trained weights for our tasks, we compared the standard Whisper-Full-Tiny model to an identical architecture with randomly



(a) Scale invariant signal to distortion ratio improvement measurement



(b) Heart rate and breathing rate algorithmic improvement measurement



(c) Heart segmentation algorithm metric measurement

Fig. 7: Performance of the model vs the number of heart/lung sound pairs used. The red line is the average across the metrics.

initialised weights. Table X shows the signal distortion results of this experiment.

For heart sound separation, the choice of weight initialisation had a negligible effect on performance. However, for lung sound separation, the absence of pre-trained weights led to a significant performance decline. While the randomly initialised model remained effective in separating lung sounds from respiratory support noise (a performance drop of only 1 dB), it struggled significantly with the "No Noise" and "General Noise" conditions.

Interestingly, the cases removed in the training set (STMV noise, a subset of General Noise cases and No-Noise cases) were the cases where the random model underperforms. This

TABLE X: Performance comparison of freezing Whisper's pre-trained weights vs freezing random weights.

Heart	No Noise		General Noise		Resp Support Noise	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
Whisper Tiny	**14.87**	**14.09**	16.67	**16.18**	**9.37**	**9.44**
+ Random Init	14.79	13.11	**16.88**	15.87	9.02	9.18

Lung	No Noise		General Noise		Resp Support Noise	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
Whisper Tiny	**10.55**	**8.81**	**10.19**	**9.97**	**14.21**	**13.52**
+ Random Init	7.55	6.70	5.14	5.47	12.96	12.78

suggests that the pretrained weights play a vital role in the generalisation of the model, where the weights provide some prior knowledge for separating lung sounds.

While the result of heart sound separation may seem surprising, the result still aligns with previous works on reprogramming random networks [5], [29]. This showcases the descriptive power of random networks. However, [30] demonstrates that reprogramming a random network does not always work. Our random model, while simultaneously performing well for separating heart sounds, failed in separating lung sounds, despite sharing the exact randomly initialised backbone, suggesting that there is more to identifying the success of reprogramming of random networks.

E. Fine-Tuning Whisper-Full

To investigate the optimality of the pre-trained embeddings via freezing, we also study the effects of fine-tuning Whispers parameters rather than freezing it. In particular, we compare the performance of fine-tuning the embeddings of the Whisper-Full models rather than freezing it as we initially assumed that the embeddings being used is sub-optimal, and fine-tuning would improve the models performance.

Here, we observed that fine-tuning Whisper-Full (e.g. 12.86 dB for Tiny, 11.74 dB for Base) offers little to no improvement compared to freezing the embeddings (e.g. 12.24 dB for Tiny, 11.59 dB for Base) when measuring average distortion improvements for both heart and lung sounds. Interestingly, fine-tuning Whisper results in a significantly lower training loss. However, this reduction does not extend to validation and testing losses, which remain comparable to those from models without fine-tuning. Furthermore, the optimal fine-tuned model was identified early in training, as overfitting became apparent in later epochs. These observations suggest that freezing the embeddings acts as a regularisation technique, preventing the model from overfitting to the smaller dataset, especially within the embedding space. This implies that the pre-trained embeddings are already effective feature extractors for most of the tasks, inherently capturing information crucial for good generalisation.

Despite these benefits, the discrepancy between the significantly lower training loss and the unchanged validation and testing losses indicates that the model, even with frozen embeddings, possesses untapped learning capacity, suggesting that the training dataset may be suboptimal in terms of size or diversity. Therefore, the proposed method of fine-tuning with

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frozen embeddings is likely most beneficial in data-limited scenarios, where the regularisation effect from frozen embeddings can enhance generalisation. Conversely, with larger and more diverse datasets, this approach may be less advantageous, as it may not fully leverage the model's learning potential, especially for cross-domain reprogramming, such as the proposed method.

F. Text Decoder Prompt Length

When reprogramming Whisper-Full, we initialised the initial prompt with {`|startoftranscript|`, `|en|`, `|transcribe|`, `|notimestamp|`}. We allowed these embeddings to be tunable, assuming the prompt was not optimal for chest sound separation. However, it is also unknown whether using four tokens is the optimal approach. Table XI investigates if we increase the number of prefix tokens from four (baseline) to sixteen (increased embedding length) for Whisper-Full-Tiny.

TABLE XI: Changes in signal distortion metrics attributed to prompt length of 4 and 16 tokens.

Heart	No Noise		General Noise		Resp Support Noise	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
4-Tokens	14.87	**14.09**	16.67	**16.18**	**9.37**	**9.44**
16-Tokens	**15.07**	14.07	**16.77**	16.11	9.11	9.01

Lung	No Noise		General Noise		Resp Support Noise	
	SDR	SI-SDR	SDR	SI-SDR	SDR	SI-SDR
4-Embeddings	10.55	8.81	10.19	9.97	14.21	13.52
16-Embeddings	**11.90**	**10.75**	**10.83**	**10.58**	**14.76**	**14.12**

Results show that increasing the prefix embedding marginally improves the signal distortion performance, with the SI-SDR improvement for no-noise lung sounds having the most significant impact. However, this may be partly due to run-to-run variance. We also do not observe any improvements in downstream tasks, with slight performance regression for breathing rate estimation. This result contrasts with [15], where, for smaller language models, increasing the prompt length generally yields a significant performance boost. However, we note that our whole prompt passed to the text decoder is tunable due to the A2E layer, while [15] only tunes the prefix embeddings. The effective prompt length could then be the prefix embeddings and those generated by the A2E layer, making the performance gain marginal, as [15] also shows that diminishing returns occur for longer prompt lengths.

VI. Limitations and Future Works

A. Diagnostic Limitation - Heart Murmur Detection

Section IV-C and Section IV-D demonstrate that performing chest sound separation enhances the performance of the heart rate and breathing rate estimation algorithms and heart segmentation algorithms. However, we would caution the use of the proposed system as a general heart sound analyser. To demonstrate this, we also investigated the impact of our separation algorithm on murmur classification using the CirCor DigiScope dataset.

We evaluated the effect of sound separation on murmur detection using the top-performing model from the PhysioNet 2022 challenge by McDonald *et al.* [31]. On the noisy CirCor DigiScope dataset, the model achieved an accuracy of 66.1%. However, after applying our separation algorithm, the accuracy for murmur detection plummeted to 7.1%. We suspect this is because our separation algorithm was not trained on sounds containing murmurs, causing it to treat murmurs as noise and remove them from the heart sound signal.

Simplifying heart sounds to only the S1/S2 components offers a key advantage for algorithms that rely on this assumption, such as those for heart rate estimation and segmentation. However, this simplification is not suitable for detecting pathological sounds, such as heart murmurs. Since our model was trained to generate only clean S1/S2 sounds, it would need to be extended to separate pathological sounds from noise for murmur detection tasks. Therefore, the current system can be viewed as a specialised structural analyser rather than a general heart sound analyser. A potential solution is to model pathological sounds as a separate channel alongside clean S1/S2 sounds. While this approach would preserve pathological sounds for detection, its implementation is beyond the scope of the current work and requires a suitable dataset. Nevertheless, the flexible architecture of our model makes it readily adaptable to this scenario if such a dataset becomes available.

B. Robustness to General Noise

While our results demonstrate the efficacy of reprogrammed Whisper models, they do not consistently outperform traditional methods like NMF under general noise conditions, suggesting that further work is needed to fully capture the complexity of diverse clinical acoustic environments.

C. Clinical Applications

While our proposed methods demonstrated promising results in the selected few real-world tests, they are currently not yet ready for direct translation into clinical settings. The underlying mechanisms of the approach require further investigation to ensure a thorough understanding of its behaviour under diverse and dynamic clinical conditions. Additionally, extensive real-world testing on external datasets is essential to validate the method's reliability, safety, and effectiveness in actual medical scenarios. Until such studies are conducted and the proposed method is rigorously evaluated against established clinical standards, its use in clinical applications remains premature.

D. Evaluations on Other Biomedical Domains

While our proposed method showed promising results in the selected biomedical domain, its generalizability remains to be further investigated. Specifically, the current evaluation has been confined to neonatal chest sound separation, which limits the broader applicability of our findings. As such, future works should validate the proposed method across a diverse set of biomedical applications to ensure its robustness and adaptability in varied real-world scenarios.

VII. Conclusion

This paper demonstrates the feasibility of reprogramming an automatic speech recognition model to perform neonatal chest sound separation using simple linear layers as input/output adapters without modifying the reprogrammed model's parameters. We have demonstrated algorithmic improvements for various downstream tasks, including heart and breathing rate estimation, as well as heart segmentation algorithms, yielding performance comparable to previous state-of-the-art separation methods.

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Appendix I
Training Setup

The models were trained on an NVIDIA RTX 4070 Ti Super paired with an Intel Core i5-14600K CPU running on Ubuntu 24.04. Table XII shows the training hyperparameters used.

TABLE XII: Training Hyperparameters

Hyperparameters	Value
Batch Size	32
Learning Rate Scheduler	Exponential Decay
Optimiser	AdamW
Max grad norm	1.0
Weight Decay	0.1
Initial Learning Rate	3×10^{-4}
Training Objective	Maximise SI-SDR